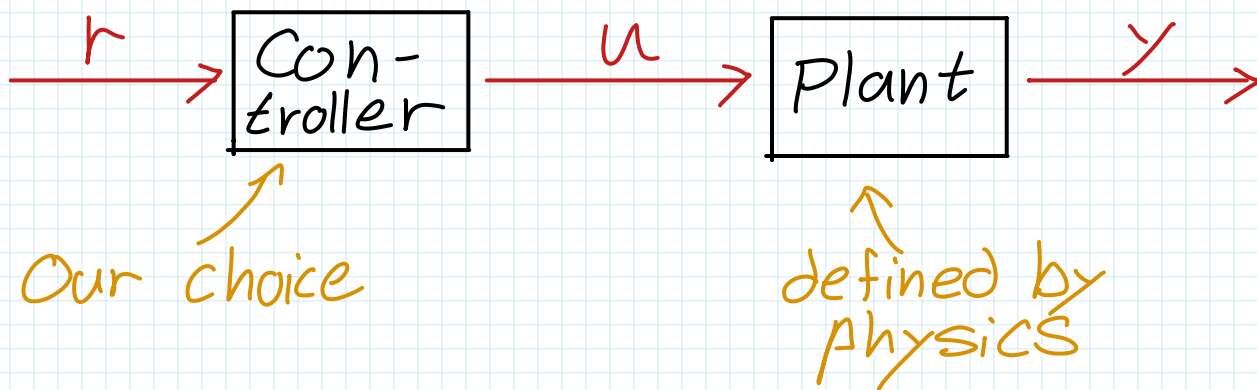


Where did we leave off?



r : Reference

u : Input

y : Output

⇒ To find a controller, we must first find out how the plant works!

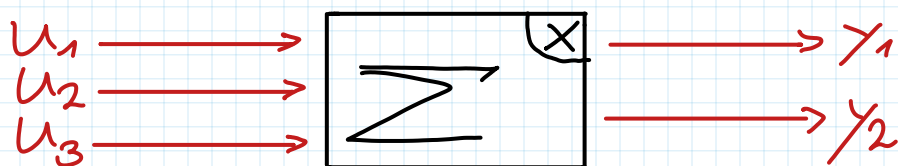
⇒ Make a model of it!

So how do we find this model?

⇒ Use equations to describe the dynamical behaviour of a system.

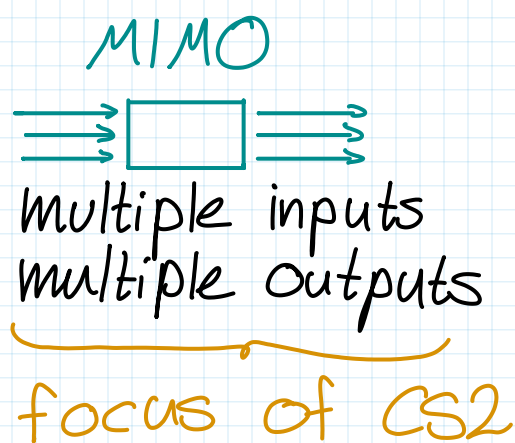
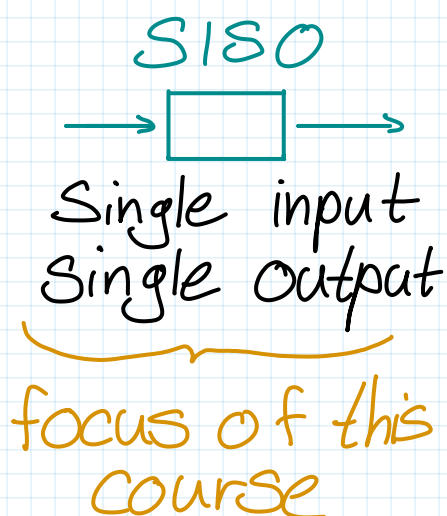
- system evolves over time
- system has inputs and outputs

The general system



- $u_i \hat{=}$ inputs
- $y_i \hat{=}$ outputs
- $\Sigma \hat{=}$ diff' eqn's of the system
- $x \hat{=}$ state variables

• Note: The general system can have multiple in-/outputs, and the #inputs can differ from the #outputs

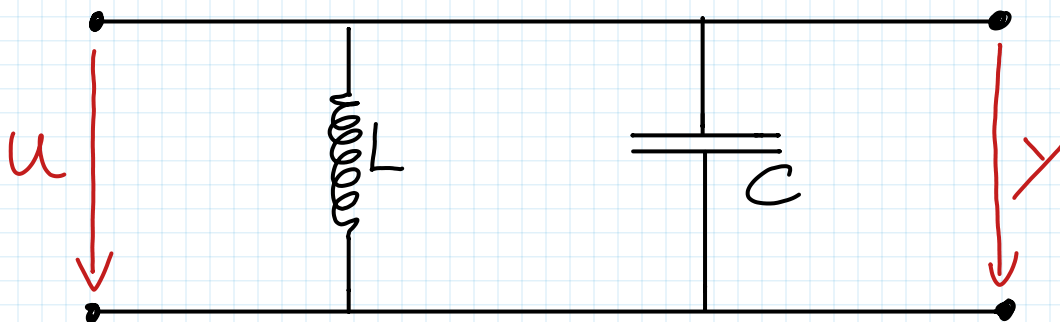


But what are state variables x ?

Def: Internal variable quantity. The output is affected by the states. They are not always directly visible if you only look at the input and output.

Example: RC - Circuit

- u : input voltage [V]
- y : output voltage [V]
- L : inductor [H]
- C : capacitor [F]



$\Rightarrow$ Output y is a function of the E -field at the capacitor and the B -field at the inductor (and the input u).

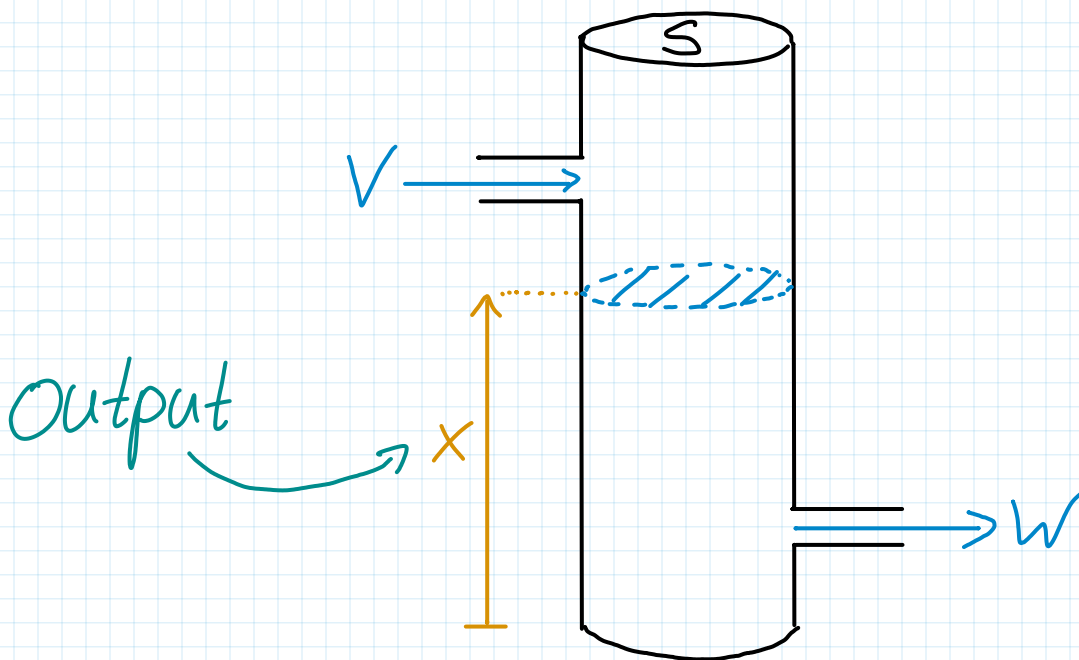
$$y = f(\underbrace{E, B}_{\text{states}}, u)$$

• We do not see the states in the output, but the output is influenced by them!

- In many cases, the output is just one of the states, which simplifies the calculation.
- Let's look at an example from the lecture.

Example: Water tank

V : inflow [m^3/s]
 W : outflow [m^3/s]
 x : height [m]
 S : area [m^2]



• Now we can use a common modeling technique called the reservoir based approach:

$$\frac{d}{dt} \text{storage} = \sum \text{inflows} - \sum \text{outflows}$$

Think of the derivative as a "change"

Here: Storage $\hat{=}$ water volume $\hat{=}$ S · X

$$\frac{d}{dt}(S \cdot X) = V - W$$

$$Y = X$$

• Simplifying:

$$\dot{X} = \frac{1}{S}(V - W)$$

$$Y = X$$

$\Rightarrow$ Any model will take this form!

• General form of diff' eqns:

$$\boxed{\begin{aligned}\dot{x}(t) &= f(x(t), u(t)) \\ y(t) &= h(x(t), u(t))\end{aligned}} \quad \begin{array}{l} \text{(I)} \\ \text{(II)} \end{array}$$

(I): Internal system dynamics

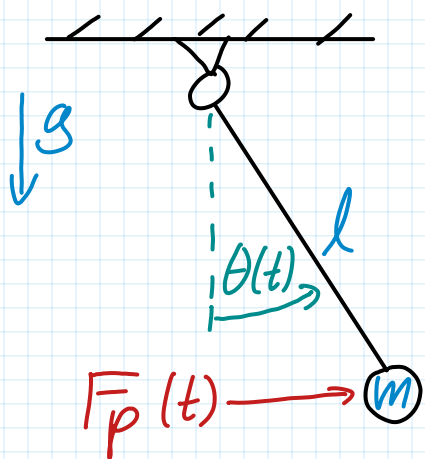
(II): System observation

Note: For a given system the eqns are not unique! We can measure multiple things on one system!
Ex: Car output $\hat{=}$ $v, \dot{v}, \dots$

But the notation above only allows for first order derivatives... This can't describe any system right!

$\Rightarrow$ True! But theres a trick.

Example: pendulum



- $l, m, g \hat{=}$ constants
- $F_p \hat{=}$ Input
- $\theta \hat{=}$ Output

1) Describe dynamics
 $\Rightarrow$ Conservation of angular momentum

$$J \ddot{\theta} = \sum M$$

$\nearrow$
Inertia
 $\uparrow$
Torques

$$l^2 \ddot{\theta}(t) = -lmg \sin(\theta(t)) + l \cos(\theta(t)) F_p(t)$$

$$y(t) = \theta(t)$$

How do we
 achieve this
 form?

$$\begin{aligned} \dot{x}(t) &= f(x(t), u(t)) \\ y(t) &= h(x(t), u(t)) \end{aligned}$$

2) Introduce new state

- Rename: $\Theta(t) = \Theta_1(t)$

- Define: $\dot{\Theta}(t) = \Theta_2(t)$

$\Rightarrow$ We deduce: $\dot{\Theta}_1(t) = \Theta_2(t)$

3) Write total system

$$\dot{\Theta}_1(t) = \Theta_2(t)$$

$$m\ell \dot{\Theta}_2(t) = -\ell m g \sin(\Theta_1(t)) + \ell \cos(\Theta_1(t)) F_p(t)$$

$$y(t) = \Theta_1(t)$$

- $\Theta_1(t), \Theta_2(t)$ are the states $x_i(t)$

- $F_p(t)$ is the input $u(t)$

- $y(t)$ is the output $y(t)$

4) Write in general form

$$\dot{x}_1(t) = x_2(t)$$

$$\dot{x}_2(t) = -\frac{g}{\ell} \sin(x_1(t)) + \frac{1}{m\ell} \cos(x_1(t)) u(t)$$

$$y(t) = x_1(t)$$

· We have achieved this form!

$$\begin{aligned}\dot{x}(t) &= f(x(t), u(t)) \\ y(t) &= h(x(t), u(t))\end{aligned}$$

Intermediate Quiz questions

Q1) Conservation of angular momentum is a reservoir based modeling approach.

☒ TRUE ☐ FALSE

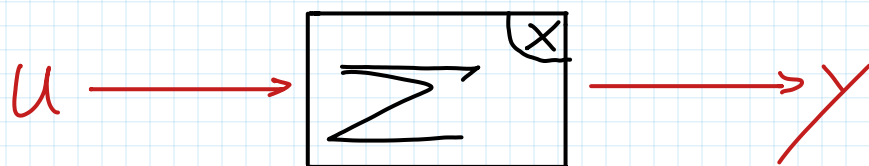
Q2) Inputs and Outputs are time dependent, but a general system is not.

☐ TRUE ☒ FALSE

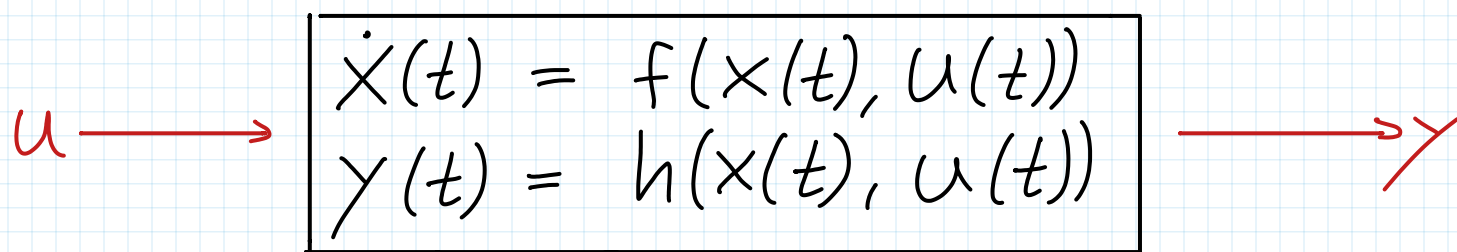
Q3) The output of a system is independent of the chosen sensor.

☐ TRUE ☒ FALSE

How does this fit into the bigger picture?



We now know what this truly is:

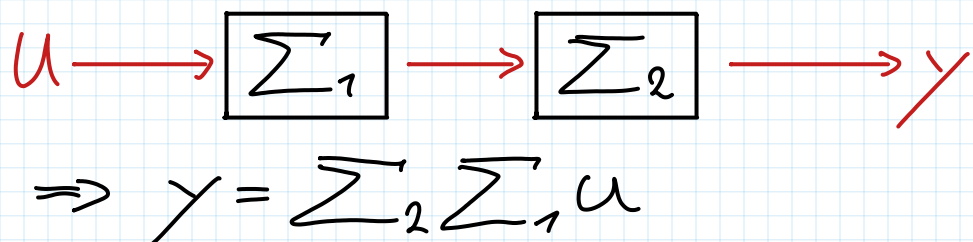


- Next week we will see what it means to linearize a system. Once we do that, we can write the signal-system relationship as such:

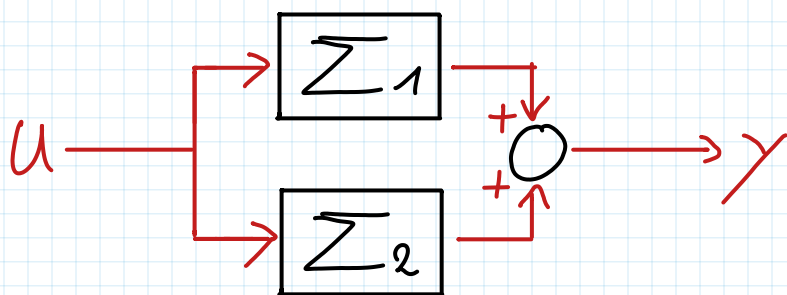
$$y = \Sigma \cdot u$$

- This will allow us to analyze more complex interconnections of systems.
- In essence, any interconnection can be broken down in three fundamental connection types.

1) Serial interconnection

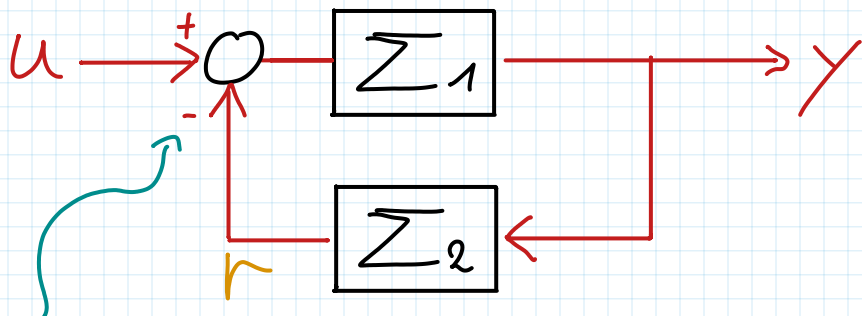


2) Parallel interconnection



$$\Rightarrow y = (\Sigma_1 + \Sigma_2)u$$

3) Feedback interconnection



Note: This defines if we have negative or positive feedback.

$$\Rightarrow y = \Sigma_1(u - r) \quad (I)$$

$$r = \Sigma_2 y \quad (II)$$

• Combining (I) and (II)

$$\begin{aligned} y &= \Sigma_1(u - \Sigma_2 y) \\ &= \Sigma_1 u - \Sigma_1 \Sigma_2 y \end{aligned}$$

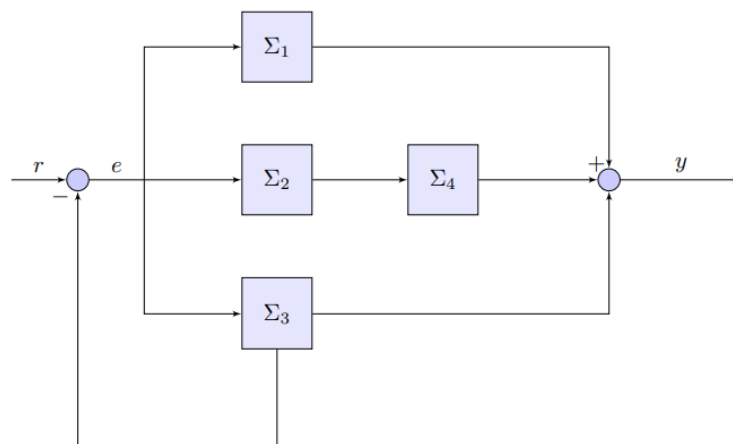
• Solve for y

$$y(1 + \Sigma_1 \Sigma_2) = \Sigma_1 u$$

$$y = (1 + \Sigma_1 \Sigma_2)^{-1} \Sigma_1 u$$

• Exam 2018)

Question 5 Choose the correct answer. (1 Point)



You are given the above system diagram. What is the associated transfer function from $r \rightarrow y$.

☐ A $\Sigma_{r \rightarrow y} = (\Sigma_1 + \Sigma_2 \Sigma_4 + \Sigma_3)$

☐ B $\Sigma_{r \rightarrow y} = \frac{(\Sigma_1 + \Sigma_2 \Sigma_4 + \Sigma_3)}{1 + \Sigma_2 \Sigma_4}$

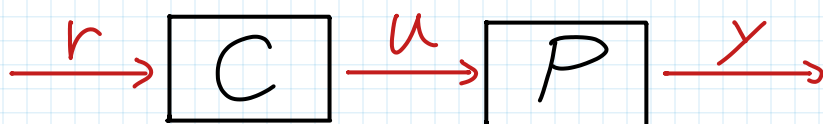
☐ C $\Sigma_{r \rightarrow y} = \frac{(\Sigma_1 + \Sigma_2 \Sigma_4 + \Sigma_3)}{1 + \Sigma_1 + \Sigma_2 \Sigma_4 + \Sigma_3}$

☒ D $\Sigma_{r \rightarrow y} = \frac{(\Sigma_1 + \Sigma_2 \Sigma_4 + \Sigma_3)}{1 + \Sigma_3}$

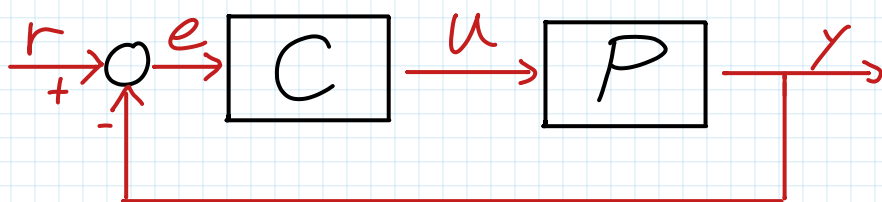
How do we now apply this to the structure we initially wanted to control?

• We had the following signals/systems:

- r : reference
- u : input
- y : Output
- C : Controller
- P : Plant



$$\Rightarrow \left. \begin{array}{l} y = P \cdot u \\ u = C \cdot r \end{array} \right\} \underline{y = PCr} \quad \left| \begin{array}{l} \text{Feed-} \\ \text{Forward} \end{array} \right.$$



$$\Rightarrow \left. \begin{array}{l} y = Pu \\ u = Ce \\ e = r - y \end{array} \right\} u = C(r - y) \left\} y = PC(r - y)$$

$$\Rightarrow y(1 + PC) = PCr$$

$$\Rightarrow y = (1 + PC)^{-1} PCr$$

$\left| \begin{array}{l} \text{Feed-} \\ \text{Back} \end{array} \right.$

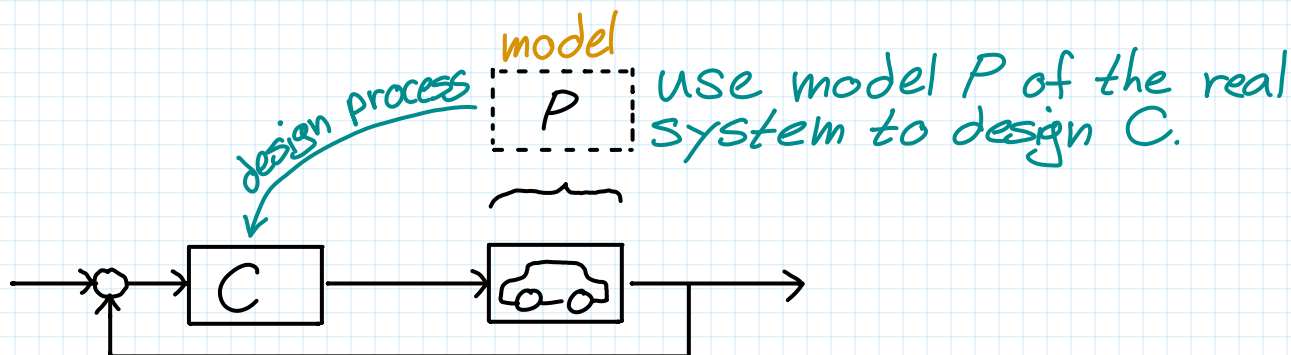
Why is feedback so crucial?

=> Feedforward only works if we have perfect knowledge of the plant in a perfect world.

- With feedback, we can:
 - + stabilize an unstable system
 - + compensate modeling uncertainties
 - + reject external disturbances
- But feedback can also:
 - introduce instability even in an otherwise stable system
 - feed noise into the system

Before we get to the quiz-questions, one final sidenote:

- What is the relation between Plant and model?



Q4) It is possible and a good idea to combine feedback and feedforward.

☒ TRUE ☐ FALSE

Q5) We treat system blockcharts just like resistances in a circuit.

☐ TRUE ☒ FALSE

Q6) When transforming a general system to a first order representation, we inevitably loose information.

☐ TRUE ☒ FALSE

Thanks for coming, have a wonderful weekend! ☺